Hierarchical ReLU-LoRA Spawning

Parameter-Efficient MoE Adaptation under Multi-Domain Conflict

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Outline

1 · Problem

2 · Method

3 · Experiments

4 · Results

5 · Summary

1 · Problem

The Core Problem: Parameter Entanglement

Setting: Fine-tuning Phi-3.5-MoE (7.6B, 16 experts, Top-2) on mixed Python + Medical data with standard LoRA.

What fails:

- Each expert gets one shared low-rank adapter $L_{k,0}$
- Both domains route tokens to the same expert $\Rightarrow$ both accumulate gradients in the same A, B matrices
- Update that helps code *partially undoes* the medical update — and vice versa

Formal condition (anti-aligned gradients):

$$\cos\big(\nabla_{\mathcal{D}_1} L_k,\ \nabla_{\mathcal{D}_2} L_k\big) < 0$$

Pre-experimental Jaccard diagnostic

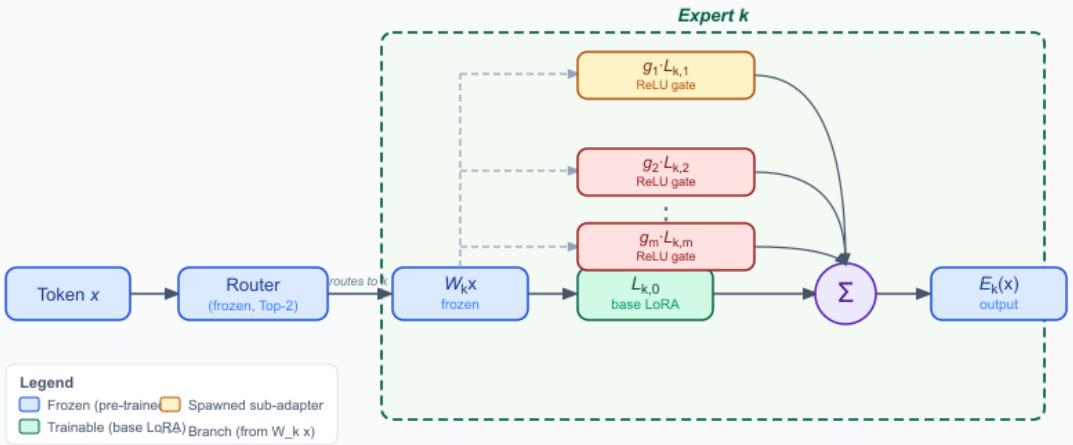
Python vs Medical: **Jaccard = 0.056** ($\approx$1 shared expert/layer). Router already separates domains. Entanglement is a **gradient-space** property of adapter architecture — not a routing problem.

2 · Method

Architecture: Spawning Sub-Adapters Inside Experts

Forward pass for a token routed to expert k:

$$E_k(x) = \underbrace{W_k x}_{\text{frozen}} + \underbrace{L_{k,0}(x)}_{\text{base LoRA}} + \sum_{j=1}^{m} \underbrace{\mathrm{ReLU}(w_{k,j}^{\top} x)}_{\text{scalar gate}} \cdot L_{k,j}(x)$$



Key properties:

- Sub-adapters structurally independent
- Gating entirely within expert k
- Global Top-K softmax **untouched**
- m grows dynamically from 0

Why not add global experts?

Softmax renormalisation drops every existing expert's weight; Top-2/16 routing causes expert dropout and loss spikes.

Bivariate Saturation Trigger

Spawning fires only when **both** conditions hold over a $W = 15$ step window:

Condition 1 — Learning Plateau

OLS slope of rank importance $< \tau = 10^{-4}$

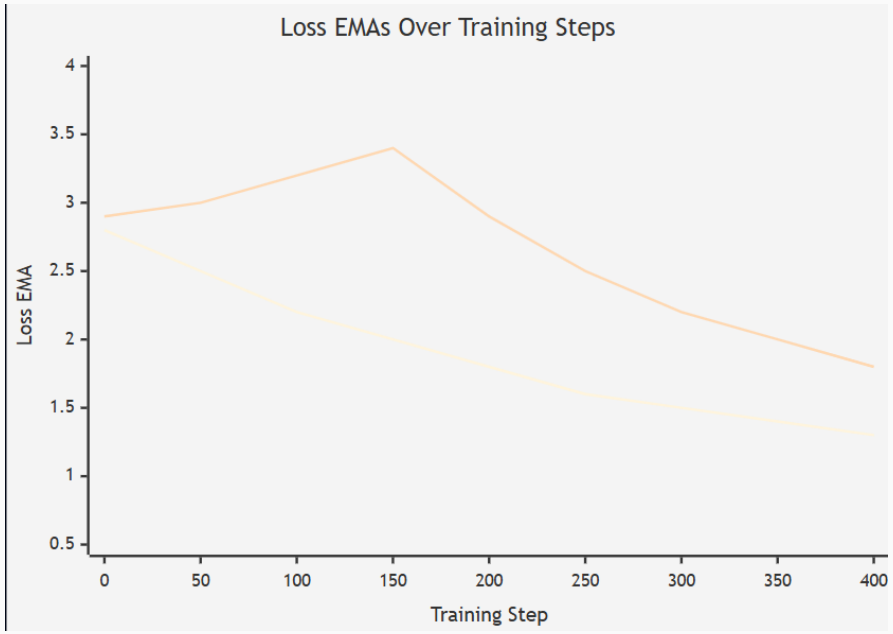
$$g_k^{(t)} = \text{mean}_r\big(\|B_{:,r}\| \cdot \|A_{r,:}\|\big)$$

Near-zero slope: base LoRA has stopped learning.

Condition 2 — Domain Conflict

$$|\mathrm{EMA}_{\mathrm{med}} - \mathrm{EMA}_{\mathrm{code}}| > \delta = 1.0$$

Genuine divergence required; calibrated to $\approx$1 spawn per 100 steps.



Sub-Adapter Initialization — Three Guarantees

1. **Zero-loss spawn** $|\Delta\text{Loss}| = 0.000000$ (verified)
   $B_{k,j} = \mathbf{0} \Rightarrow L_{k,j}(x) = 0$ at construction. Loss mathematically unchanged.

2. **Gradient-informed A init**
   $A \leftarrow$ top-r rows of V^H from SVD of residual gradient $G = \nabla_B^\top \nabla_A$. Orients new adapter toward directions the base adapter could not learn.

3. **Symmetry-breaking gate noise** gate grad $= 6.99 \times 10^{-3}$ at step+2
   $w_{k,j} \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 10^{-3} \cdot \text{Var}(W_k)$. Sufficient to escape the symmetric saddle; gate receives non-zero gradient within 2 steps of spawn.

3 · Experiments

Experimental Setup

Model & target

- Phi-3.5-MoE-instruct, 7.6B params
- Single expert, Layer 0 (highest Jaccard overlap)
- bf16, AdamW, lr$= 2 \times 10^{-5}$, base rank 16

Three methods

- Standard LoRA (spawning off)
- DR-LoRA (rank doubles at step 750)
- **Hierarchical Spawning** (full)

Conflict-scaling grid (1500 steps each):

Run	Conflict	Composition
A	0%	164 code / 0 medical
B	20%	164 code / 310 medical
C	50%	164 code / 775 medical

Metrics: PPL_{code}, Negative Transfer, Sub-router JSD

4 · Results

Results — Negative Transfer Across Conflict Ratios

Method	Run	PPL_{code}	Neg. Transfer	Spawns	JSD
LoRA	A (0%)	93.9	—	0	—
LoRA	B (20%)	192.3	+98.4	0	—
LoRA	C (50%)	1155.4	+1061.5	0	—
DR-LoRA	A	57.7	—	1	—
DR-LoRA	B	212.2	+154.4	1	—
DR-LoRA	C	474.1	+416.4	1	—
Hier.	A	58.7	—	0	—
Hier.	B	147.2	**+88.5**✓	10	0.154
Hier.	C	332.3	**+273.6**✓	10	0.122

Eager-Spawn Ablation — What Is the Binding Constraint?

Hypothesis: Is late trigger firing the bottleneck at 50% conflict?

Configuration	First spawn	Spawns	PPL	Neg. Transfer
Standard (δ=1.0, cap 10)	Step 155	10	332.3	+273.6
Eager (δ=0.5, cap 15)	Step 153	15	868.0	+774.1

Two findings

1. Timing was **never** the issue — halving δ shifts first spawn by only 2 steps
2. **Cap is the binding constraint:** 15 sub-adapters spreads gradients too thinly; no adapter develops strong domain specificity

Next: conflict-ratio-aware dynamic cap

$$\mathrm{cap}(t) = \mathrm{round}\Big(5 + 10 \cdot \tanh\big(|\mathrm{EMA}_{\mathrm{med}} - \mathrm{EMA}_{\mathrm{code}}|\big)\Big)$$

5 · Summary

Summary

1. **Diagnostic first:** Jaccard $= 0.056$ proves entanglement is gradient-space, not routing-space
2. **Architecture:** Independent ReLU-gated sub-adapters spawned inside saturated experts; global router untouched
3. **Trigger:** Bivariate monitor (plateau $\wedge$ conflict) ensures data-driven spawning
4. **Stability:** Zero-loss spawn $+$ gradient-informed A init $+$ symmetry-breaking gate noise
5. **Results:** Best negative transfer at 20% and 50% conflict; JSD confirms structural gate specialisation
6. **Ablation:** Cap (not timing) is binding constraint $\Rightarrow$ dynamic cap is next step